

Electrical Instrumentation Part-II

PowerPack Phase 1: CRO and Oscilloscope Mastery

Reverse-Engineering Roadmap for Semester Answer Writing

Purpose of this PowerPack

This Phase 1 pack is not ordinary reading material. It is a reverse-engineered exam weapon for the CRO part of Electrical Instrumentation Part-II. After finishing this pack, the target is not merely to remember definitions. The target is to look at any CRO question and immediately know:

1. which diagram to draw,
2. which principle to write first,
3. which formula to derive,
4. which final line gives marks,
5. which mistakes must be avoided in the semester script.

Strict diagram command for every answer

Whenever a diagram is required, follow these rules in the answer script:

1. Draw the diagram before the long explanation.
2. Keep blocks rectangular and wires horizontal or vertical as far as possible.
3. Never write labels on top of wires or components.
4. If two labels are close, put one above and one below, or one left and one right.
5. Long labels must be split into two or three lines.
6. For CRO waveforms and Lissajous figures, curves are allowed, but all guide lines must be straight and clean.

Contents

1 Phase 1 Attack Plan

What belongs to Phase 1?

Phase 1 covers all CRO and waveform-display type questions. In the last papers, these appear repeatedly as small and medium questions. They are high-return topics because the same diagrams and derivations are reused.

1.1 Question Families

Question family	How to recognize it	Main weapon
CRO functional blocks	“Elucidate functional blocks”, “brief details of CRO”	CRO block diagram + block functions
CRT construction	“Cathode ray tube”, “electron gun”, “deflection plates”	CRT internal diagram
Time-base generator	“steady horizontal sweep”, “saw-tooth”, “wave generating circuit”	Saw-tooth waveform + RC sweep circuit
Trigger input	“trigger input”, “stable waveform”, “synchronization”	Trigger block + level/slope crossing
Delay line	“delay line in vertical deflection system”	vertical signal delayed so leading edge appears
Matched probe	“probe compensation”, “matched capacitance”, “necessity of matched probe”	compensated divider formula
Lissajous phase	“phase difference between same frequency signals”	ellipse + $\sin \phi = y_1/y_2$
Lissajous frequency	“horizontal tangencies”, “vertical tangencies”	$f_y/f_x = N_H/N_V$

1.2 Two-Day Use of Phase 1

Minimum winning schedule

1. **First 45 minutes:** learn the CRO block diagram and the CRT internal diagram.
2. **Next 45 minutes:** learn time-base, trigger, delay line.
3. **Next 45 minutes:** learn probe compensation and Lissajous phase derivation.
4. **Next 30 minutes:** write all final formulae from memory.
5. **Last 30 minutes:** draw all diagrams once without seeing.

1.3 Answer-Writing Skeleton

For almost every CRO question, write in this order:

1. **Definition or principle.**
2. **Clean labelled diagram.**
3. **Working block-by-block.**
4. **Formula or derivation.**
5. **Importance / final conclusion.**

2 CRO From Absolute Zero

2.1 Basic idea

A Cathode Ray Oscilloscope is an instrument that converts an electrical signal into a visible waveform. It does this by moving a bright spot on a fluorescent screen. The vertical movement represents voltage and the horizontal movement represents time.

$$\text{Vertical deflection} \propto \text{input voltage}$$

$$\text{Horizontal deflection} \propto \text{time}$$

One-line exam definition

A CRO is an electronic measuring instrument used to display and measure voltage waveforms with respect to time by using an electron beam, deflection plates and a fluorescent screen.

2.2 Functional Block Diagram of CRO

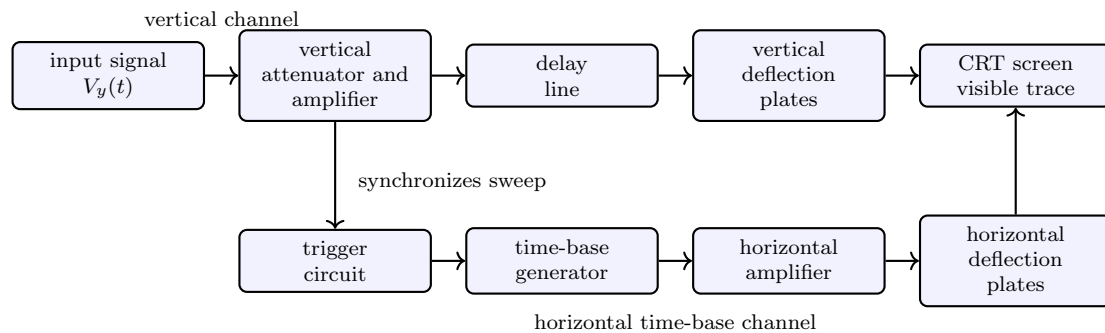


Figure 1: Functional block diagram of a Cathode Ray Oscilloscope.

2.3 Block-by-block function

1. **Vertical attenuator and amplifier:** adjusts and amplifies the input signal so that it can produce proper vertical deflection.
2. **Delay line:** delays the vertical signal slightly so that the beginning of the waveform is visible after the trigger starts the sweep.
3. **Trigger circuit:** starts each sweep at the same point of the waveform, giving a steady display.
4. **Time-base generator:** produces saw-tooth voltage for horizontal deflection.
5. **Horizontal amplifier:** amplifies the sweep voltage.
6. **CRT:** produces, deflects and displays the electron beam.

Common mistake

Do not write that the vertical amplifier produces horizontal sweep. The vertical amplifier only controls up-down motion. The time-base generator and horizontal amplifier control left-right motion.

3 CRT Construction and Working

3.1 Exam definition

A Cathode Ray Tube is the display tube of a CRO. It produces a beam of electrons, accelerates and focuses it, deflects it using electric fields, and finally makes it strike a fluorescent screen to produce a visible spot.

3.2 Internal Diagram of CRT

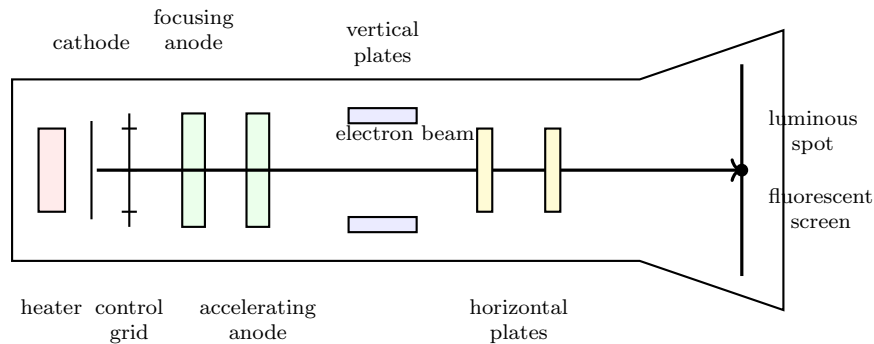


Figure 2: Simplified internal construction of a Cathode Ray Tube.

3.3 Working in exam language

The heater warms the cathode. The cathode emits electrons by thermionic emission. The control grid controls the intensity of the electron beam. The focusing anode makes the beam narrow and sharp. The accelerating anode gives high velocity to the electrons. The vertical and horizontal plates deflect the beam according to the applied voltages. When the beam strikes the fluorescent screen, a bright spot is produced. By moving this spot in two directions, the waveform is traced.

4 Time-Base Generator and Saw-Tooth Sweep

4.1 Why saw-tooth voltage is necessary

In CRO, the vertical plates receive the input signal. Therefore, vertical motion gives signal amplitude. To display amplitude versus time, the electron beam must also move horizontally with uniform speed.

For this, a saw-tooth voltage is applied to the horizontal plates.

$$V_x(t) = kt$$

Since horizontal deflection is proportional to horizontal voltage,

$$x \propto V_x(t)$$

$$x \propto kt$$

$$\boxed{x \propto t}$$

Thus the horizontal axis becomes a time axis.

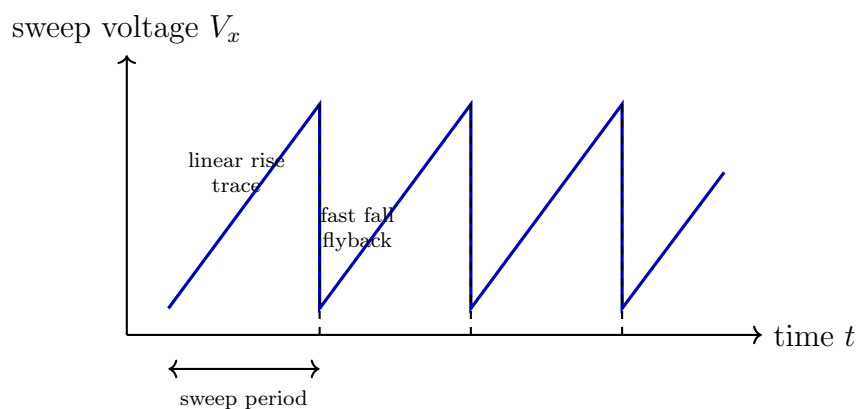


Figure 3: Saw-tooth sweep voltage for horizontal deflection.

4.2 Basic sweep generating circuit

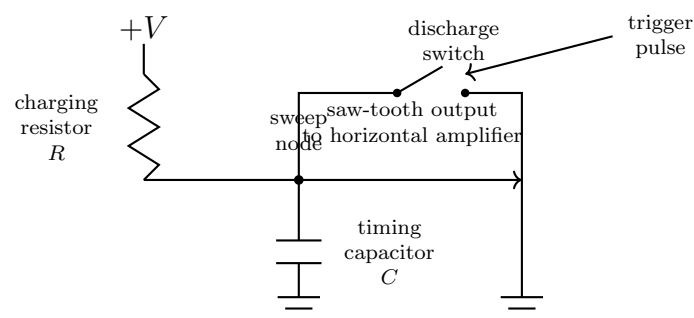


Figure 4: Basic RC sweep generator used to produce saw-tooth voltage.

4.3 Operation

1. During trace, the switch is open and capacitor C charges through R . The capacitor voltage rises almost linearly.
2. This rising voltage moves the spot from left to right.
3. At the end of the sweep, the trigger closes the discharge switch.
4. Capacitor discharges very quickly. The spot returns to the left side. This is flyback or retrace.

Final line for exam

A steady saw-tooth sweep voltage is necessary because its linear rising part makes the horizontal deflection proportional to time, while its rapid falling part brings the beam back quickly for the next trace.

5 Trigger Input and Stable Display

5.1 Definition

Trigger input is the signal input to the trigger circuit of a CRO. It controls the starting instant of the time-base sweep. Its main function is to make every sweep start from the same voltage level and same slope of the input waveform.

5.2 Trigger action diagram

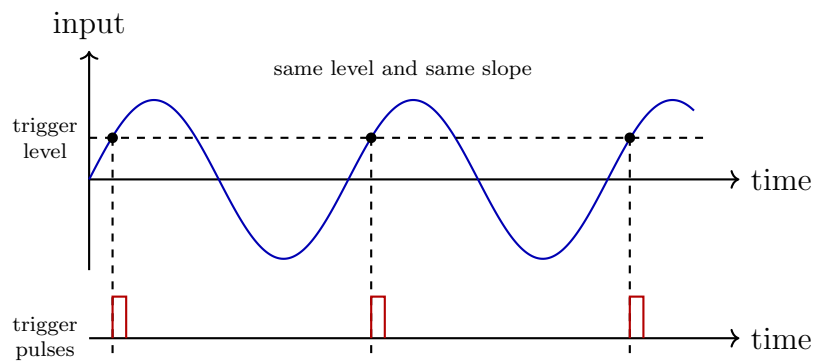


Figure 5: Trigger circuit produces a pulse whenever the waveform crosses the selected level with selected slope.

5.3 Functions

1. It synchronizes horizontal sweep with the input signal.
2. It gives stable and stationary waveform display.
3. It allows level selection and slope selection.
4. It prevents rolling or drifting of the waveform.
5. It improves measurement accuracy for time period, frequency and phase.

6 Delay Line in Vertical Deflection System

6.1 Why delay line is required

The trigger circuit and time-base generator need a small finite time to start the horizontal sweep. If the input signal reaches the vertical plates immediately, the beginning portion of the waveform may be lost before the sweep starts. To avoid this, the vertical signal is delayed slightly by a delay line.

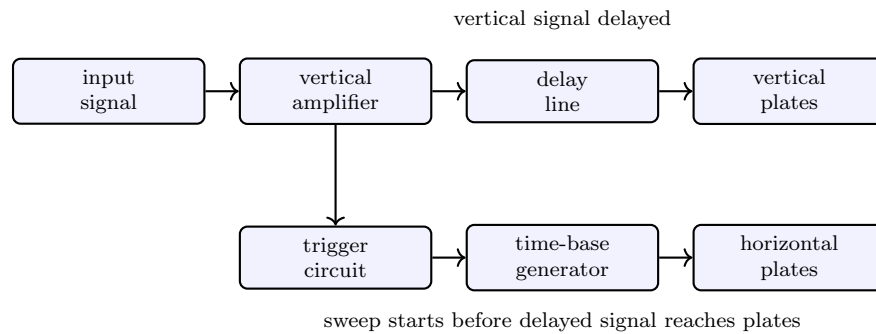


Figure 6: Role of delay line in the vertical deflection system.

Two-mark final answer

The delay line delays the vertical signal so that the horizontal sweep has enough time to start. Thus the leading edge of the input waveform is displayed on the screen instead of being missed.

7 Matched Probe and Probe Compensation

7.1 Why matched probe is necessary

A CRO input has resistance and capacitance. The probe also has resistance and capacitance. Together they form a frequency-dependent voltage divider. If the time constants are not matched, different frequency components are attenuated differently. Then square waves become rounded or overshoot and measured amplitude becomes wrong.

7.2 Equivalent circuit

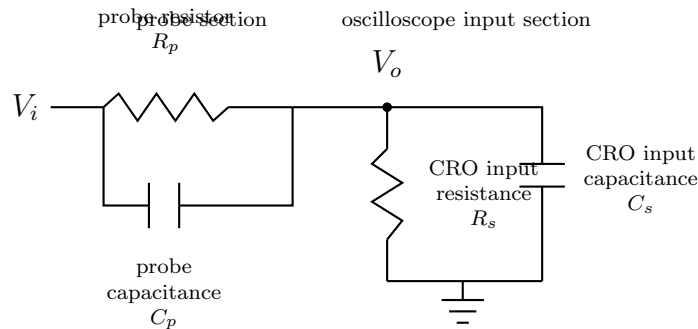


Figure 7: Equivalent circuit of a compensated CRO probe and CRO input.

7.3 Derivation of compensation condition

For proper compensation, the two time constants must be equal:

$$R_p C_p = R_s C_s$$

Equivalently, using capacitive reactance:

$$R_s X_{C_p} = R_p X_{C_s}$$

Now,

$$X_C = \frac{1}{\omega C}$$

Therefore,

$$R_s \left(\frac{1}{\omega C_p} \right) = R_p \left(\frac{1}{\omega C_s} \right)$$

Cancelling ω ,

$$\frac{R_s}{C_p} = \frac{R_p}{C_s}$$

Hence,

$$R_s C_s = R_p C_p$$

So,

$$C_p = \frac{R_s C_s}{R_p}$$

Numerical command

Before substituting, convert all units to SI units. Mega-ohm to ohm, kilo-ohm to ohm, microfarad to farad, picofarad to farad.

8 Lissajous Method for Phase Difference

8.1 Principle

When two sinusoidal signals of the same frequency are applied to the X and Y inputs of a CRO in X-Y mode, the internal time-base is switched off. The screen displays a Lissajous ellipse. From the intercepts of this ellipse, phase difference can be calculated.

8.2 Applied signals

$$x = x_0 \sin \omega t$$

$$y = y_0 \sin(\omega t + \phi)$$

where ϕ is the phase difference.

8.3 Clean diagram

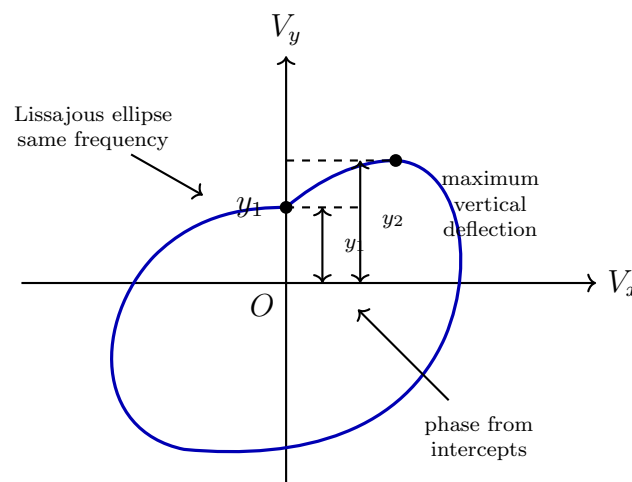


Figure 8: Lissajous ellipse for phase difference measurement.

8.4 Derivation

At the Y-axis intercept, $x = 0$. Hence,

$$x_0 \sin \omega t = 0$$

$$\sin \omega t = 0$$

$$\omega t = 0, \pi, 2\pi, \dots$$

At these instants,

$$\cos \omega t = \pm 1$$

Now,

$$y = y_0 \sin(\omega t + \phi)$$

$$y = y_0 [\sin \omega t \cos \phi + \cos \omega t \sin \phi]$$

At $x = 0$, $\sin \omega t = 0$, so

$$y = \pm y_0 \sin \phi$$

Taking magnitude,

$$y_1 = y_0 \sin \phi$$

The maximum vertical deflection is

$$y_2 = y_0$$

Therefore,

$$\frac{y_1}{y_2} = \sin \phi$$

$$\phi = \sin^{-1} \left(\frac{y_1}{y_2} \right)$$

8.5 Special cases

Phase	Observed pattern
0°	straight line with positive slope
90°	circle if amplitudes are equal, otherwise ellipse
180°	straight line with negative slope

9 Lissajous Method for Frequency Measurement

9.1 Principle

When two sinusoidal signals of different frequencies are applied to the X and Y inputs of a CRO, a stationary Lissajous figure is obtained only when the frequency ratio is a rational number. The ratio can be obtained from the number of tangencies.

9.2 Formula

Let

N_H = number of horizontal tangencies

N_V = number of vertical tangencies

Then,

$$\frac{f_y}{f_x} = \frac{N_H}{N_V}$$

where f_x is the horizontal input frequency and f_y is the vertical input frequency.

9.3 Example pattern from paper style

If horizontal tangencies are 33, vertical tangencies are 22, and horizontal frequency is 3 kHz, then

$$\frac{f_y}{3} = \frac{33}{22} = \frac{3}{2}$$

$$f_y = \frac{3}{2} \times 3 = 4.5 \text{ kHz}$$

$$f_y = 4.5 \text{ kHz}$$

Exam command

Do not confuse tangencies with frequencies. Horizontal tangencies go to the numerator for vertical frequency.

10 Question Templates to Practice

Template 1: Explain functional blocks of CRO

Write: definition, block diagram, vertical channel, horizontal channel, trigger, delay line, CRT, final answer.

Diagram required: CRO functional block diagram.

Final line: The CRO converts electrical signal variation into a visible time-domain waveform by vertical deflection proportional to voltage and horizontal deflection proportional to time.

Template 2: Explain trigger input

Write: trigger definition, level and slope, stable display, trigger pulse, time-base start.

Diagram required: trigger timing graph or block diagram.

Final line: Triggering makes every sweep begin at the same point of the waveform, giving a stationary display.

Template 3: Explain saw-tooth sweep

Write: need of time axis, $V_x = kt$, trace, flyback, RC charging circuit.

Diagram required: saw-tooth waveform and sweep generator.

Final line: Linear rise gives uniform trace and rapid fall gives fast retrace.

Template 4: Matched probe numerical

Write: compensated divider, condition $R_p C_p = R_s C_s$, substitute data, final capacitance.

Diagram required: probe-CRO equivalent circuit.

Formula:

$$C_p = \frac{R_s C_s}{R_p}$$

Template 5: Phase by Lissajous

Write: X-Y mode, same frequency, ellipse, derivation from $x = 0$, final formula.

Diagram required: ellipse with y_1 , y_2 , dashed guides.

Formula:

$$\phi = \sin^{-1} \left(\frac{y_1}{y_2} \right)$$

11 Formula Bank for Phase 1

Concept	Formula / result
Horizontal sweep	$V_x(t) = kt$, hence $x \propto t$
Probe compensation	$R_p C_p = R_s C_s$
Probe capacitance	$C_p = \frac{R_s C_s}{R_p}$
Lissajous phase	$\sin \phi = \frac{y_1}{y_2}$
Lissajous phase result	$\phi = \sin^{-1} \left(\frac{y_1}{y_2} \right)$
Alternative phase formula	$\phi = \sin^{-1} \left(\frac{x_1}{x_2} \right)$
Frequency ratio	$\frac{f_y}{f_x} = \frac{N_H}{N_V}$

12 Last-Day Self-Test

Write these without seeing

1. Draw CRO functional block diagram.
2. Draw CRT internal diagram.
3. Draw saw-tooth waveform and explain trace/flyback.
4. Draw matched probe equivalent circuit and derive $R_p C_p = R_s C_s$.
5. Draw Lissajous ellipse and derive $\phi = \sin^{-1}(y_1/y_2)$.
6. Solve one tangency frequency problem.

DSD/BJB style final command

For a semester answer, never jump directly to formula. Start from physical meaning, draw a clean labelled diagram, then derive formula slowly. Your answer must look like the examiner can award marks block-by-block even before reading every line.